WORKING TECHNICAL BRIEF · 28 SEPTEMBER 2026

Palisades Fire Level-Set PINN

How the real-world data enter the model, how the code trains the forecast, and what remains before the result is reportable.

Current state. A stable 4,000-epoch CPU run completed; total loss fell from 0.1657 at epoch 500 to 0.1192 at epoch 4,000 with no divergence or NaNs. Forecast accuracy remains pending until the four held-out boundary errors run successfully.

1 · OBSERVE VIIRS active-fire detections > **2 · ALIGN** 90 m shared raster grid > **3 · DERIVE** SDF targets + spread rate > **4 · TRAIN** Physics + observed fronts > **5 · TEST** Four unseen overpasses

Data become two kinds of model input

NASA FIRMS VIIRS 375 m detections

Fourteen overpasses from 7-12 January 2025. Spatial quality control retained 920 of 1,056 nominal/high-confidence points. Each alpha-hull boundary becomes a signed-distance field (SDF); overpasses 0-9 train the model and 10-13 remain unseen.

LANDFIRE LF2020 terrain and fuels

Elevation, slope, and FBFM40 fuel class are reprojected to the common 178 × 290 grid at 90 m in EPSG:3310. Fuel and slope become spatial multipliers in the spread-rate field.

GridMET daily weather

Wind speed, direction, and 100-hour fuel moisture cover 25 daily steps, 7-31 January. They supply wind and moisture terms in R. Native resolution is about 4 km; upsampling aligns cells but adds no local detail.

Derived products

Filtered perimeters, a 10 × 178 × 290 SDF cache, the aligned environmental stack, and daily R(x,y,t) are saved separately so data preparation and training can be reproduced.

Interpretation stays deliberately narrow

Observation limit. VIIRS hulls are thermal-detection footprints, not cumulative burned-area perimeters; their area can shrink as detections change.

Physics limit. The current R field is a calibrated first-order proxy using fuel, wind, slope, and moisture. It is not yet full Rothermel.

The code learns a moving fire front

A tanh network maps normalized **(time, x, y)** to level-set value Φ; the predicted fire boundary is Φ = 0. Automatic differentiation supplies the derivatives in the physics residual.

$$\Phi_\tau + (R_{eff} T/L)\, \|\nabla\Phi\| = 0$$

$$\text{Loss} = 10\, L_{IC} + 1\, L_{PDE} + 5\, L_{OBS}$$

1. **Initial condition:** Fit the first SDF to place the starting front.
2. **Physics:** Penalize PDE violations at sampled space-time points.
3. **Observations:** Fit nine later SDFs; preserve four chronological holdouts.
4. **Direction:** Project wind direction onto the learned front normal ($C_{WDIR} = 0.6$).

Network	3 → 128 × 4 → 1; tanh
Training	Adam, learning rate 10^{-3}, 4,000 epochs
Samples / epoch	8,192 PDE; 4,096 IC; 1,024 per observation target
Scales	T = 4.53 days; L = 26,010 m; T/L = 15.1 s/m
Observed losses	IC 0.00196; PDE 0.03646; OBS 0.01263 at epoch 4,000

Held-out distance is the decision metric

For overpasses 10-13, the code extracts Φ = 0 and reports mean predicted-to-observed boundary distance in meters. A missing contour is a failure, not zero error. Distance is preferred to area because VIIRS footprints are not cumulative burn scars.

NEXT REVISION AFTER EVALUATION RUNS

- Add the four held-out errors and their mean.
- Add one predicted-versus-observed perimeter figure.
- Record the checkpoint and exact run configuration.
- Only then decide whether to tune R scale, observation weight, or directional gain.